

Variation in Multimodal LLM Diagnostic Performance Across ICD-Stratified Clinical Domains

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Abstract. Safe clinical use of multimodal large language models (LLMs) requires evaluation beyond average diagnostic performance. We evaluated 10 frontier LLMs on a South African inpatient diagnostic dataset, grouping cases by International Classification of Diseases 10th Revision (ICD-10) chapter. We compared each model’s ICD-10 chapter-level performance with its overall performance and identified statistically significant differences across chapters. There are several explanations consistent with these data, such as LLM underperformance, variations in case complexity, documentation limitations, or systematic variation in the quality of the benchmark reference standard. Resolving this ambiguity is a key step in evaluating artificial intelligence models as clinical decision support tools.

Keywords: Artificial Intelligence, Multimodal Large Language Models, LMIC, Diagnosis, Safety, Internal Medicine, Paediatrics.

1 Introduction

As large language models (LLMs) are increasingly considered for clinical decision support, diagnostic-accuracy evaluations often still rely on datasets and clinical contexts from high-income settings [1]. To address the need for real-world low- and middle-income country (LMIC) evidence, the Validation of LLMs versus Doctors in Diagnosis (VALID) study evaluated ten multimodal LLMs using a multimodal South

African inpatient dataset [2]. The initial VALID analysis focused on aggregate performance and showed that all LLMs scored higher than ward-documented diagnoses, with performance varying by only 13% between the highest- and lowest-performing LLMs despite a ~50-fold spread in cost per case [2].

Aggregate performance may, however, obscure clinically important subgroup signals, including systematic model performance differences in particular disease categories [3]. This matters in LMIC settings, where resource limitations may reduce opportunities for review by more experienced clinicians [4], increasing the risk that harmful or misleading guidance from AI tools goes unrecognized.

We grouped VALID cases by mapping expert-assigned International Classification of Diseases 10th Revision (ICD-10) codes to their corresponding chapters. We then compared each model’s chapter-level performance to its mean performance to assess variation in chapter-level performance shared across models. We propose ICD-10 chapter stratification with model-relative comparisons as a practical, low-cost approach for subgroup evaluation in datasets that already contain ICD codes and demonstrate its application to a real-world LMIC dataset.

2 Methods

VALID was a retrospective, cross-sectional evaluation of multimodal LLM diagnostic accuracy conducted at Chris Hani Baragwanath Academic Hospital in Johannesburg, South Africa. Ethics approval for the VALID study and subsequent analyses was granted by the University of the Witwatersrand Human Research Ethics Committee (M241153 and M251102). Internal Medicine and Paediatric inpatient records were abstracted into detailed multimodal cases. Expert panels, composed of two specialist clinicians, then established reference-standard diagnoses, differential diagnoses and reasoning for these cases. A three-model LLM jury, in which LLMs served as judges, was calibrated to expert-panel scoring and scored outputs from ten LLMs from four different providers across the diagnosis, differential diagnosis, reasoning, and safety domains (scored on 1–5 Likert scales) [5]. The assessed models were selected to reflect high-performing general-purpose multimodal models at the time of analysis. Model vendors included Anthropic, Google, OpenAI, and xAI (see Fig. 1 for the complete list of models).

For each model m and ICD-10 chapter c , we calculated a model-relative chapter delta, $\Delta_{m,c} = \mu_{m,c} - \mu_m$, where $\mu_{m,c}$ is model m ’s mean score within chapter c and μ_m is the same model’s overall mean score across all included cases. This reduced the influence of baseline performance differences among models. Shared chapter-level performance uncertainty across all models was estimated using 10,000 bootstrap iterations, resampling model-case delta values with replacement within each chapter, with model overall means held fixed, 95% percentile confidence intervals, and exploratory two-sided bootstrap p-values, using an add-one finite-resampling correction to avoid reporting p-values as zero. These p-values are not corrected for multiple hypothesis testing, as this analysis serves as a screening step to identify candidate chapter-level effects.

3 Results

We analyzed 300 cases across 18 ICD-10 chapters and 10 LLMs, yielding 2734 model-case evaluations, as individual model response rates ranged from 65–100%. Diagnosis-score model-relative chapter deltas varied across ICD-10 chapters (Fig. 1). The deltas of the greatest magnitude were associated with chapters with the smallest sample sizes. To limit emphasis on estimates from small samples, figures were restricted to ICD-10 chapters with at least 5 cases. Main performance signals were limited to chapters with at least 10 cases. Additional exploratory figures are available from the corresponding author on reasonable request.

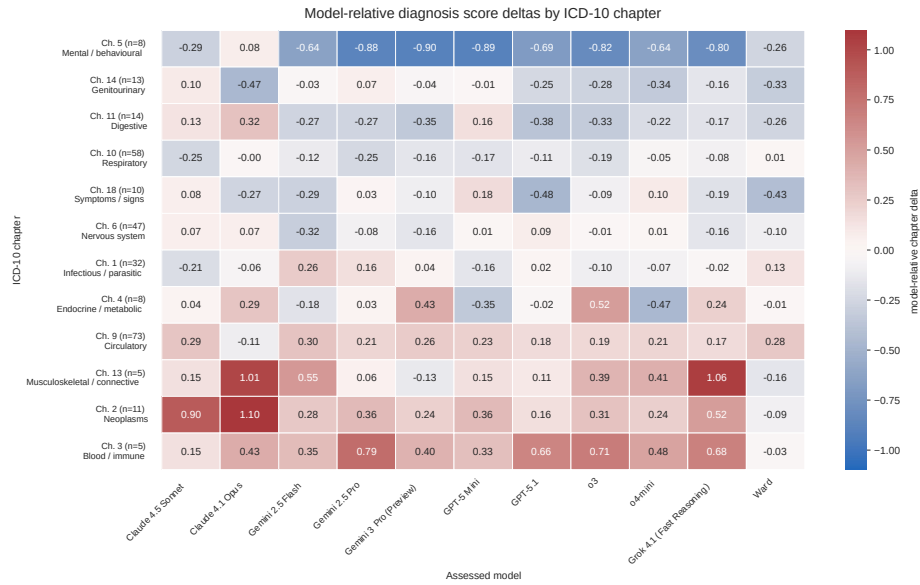


Fig. 1. Model-relative diagnosis-score deltas by ICD-10 chapter for evaluated models and ward comparator, among chapters with at least 5 cases. Cells show point estimates (chapter mean minus model overall mean) without confidence intervals; see Fig. 2 for bootstrapped uncertainty. Negative/blue and positive/red values indicate lower and higher performance relative to each model’s baseline. Row labels show chapter number, case counts and abbreviated descriptions.

Among chapters with at least 10 cases, the respiratory chapter showed diagnosis ($p = 0.0014$) and reasoning ($p = 0.0052$) score shortfalls, while digestive ($p < 0.001$) and genitourinary ($p = 0.0014$) chapters showed safety-only shortfalls - each observed across all ten models. This suggests that diagnostic accuracy alone may miss chapter-linked safety concerns. Because these findings were based on exploratory, unadjusted p -values, they should be interpreted as screening-level signals rather than confirmatory subgroup effects.

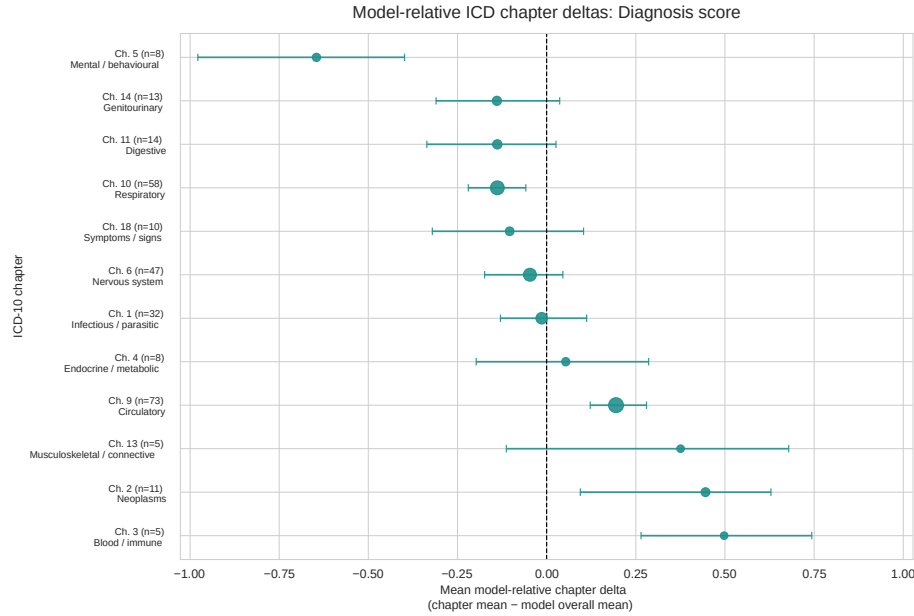


Fig. 2. Mean model-relative diagnosis-score deltas across evaluated models, by ICD-10 chapter. Only chapters with at least 5 cases were included. Points show chapter-level deltas with 95% bootstrap confidence intervals, with point size proportional to chapter case count. Row labels include case counts and abbreviated descriptions. Negative values indicate lower diagnostic performance after accounting for each model’s overall baseline.

4 Discussion

This ICD-stratified analysis identified potentially important chapter-level performance differences shared across the assessed multimodal LLMs. In this LMIC inpatient benchmark, aggregate performance summaries would not have revealed that lower safety scores concentrated in specific diagnostic categories, while diagnosis scores remained at or above model baselines.

These patterns should be interpreted cautiously, with several non-mutually exclusive explanations to consider: (1) underestimated confidence intervals potentially due to small sample sizes; (2) systematic model weaknesses by chapter; (3) chapter-dependent variation in reference-standard reliability; (4) greater case complexity in some chapters; (5) chapter-linked differences in documentation quality or completeness (e.g. missing dermatological photographs); and (6) possible chapter-related bias in the LLM jury used for scoring. Interpretation is further limited as ICD-10 chapters are broad coding categories rather than precise disease groupings.

Thorough assessment of LLMs for potential blind spots is an important step in building confidence in their use for clinical decision support, and this requires going beyond averages. Confirmatory work should apply formal False Discovery Rate (FDR) correction to distinguish genuine chapter-level effects from chance findings, and

explore linear mixed-effects models to adjust for variation in case complexity across chapters.

Local evaluations should examine whether performance shortfalls cluster by clinical subgroup, score domain, or regionally relevant case features before LLMs are considered adequately assessed. This is especially important in LMIC inpatient settings, where disease distributions, documentation quality, resource availability, and access to senior clinical oversight may differ from those in high-income benchmark settings.

Our immediate next steps include a qualitative review of signal-generating cases to distinguish model weakness from case complexity, documentation gaps, panel uncertainty, and LLM-jury effects. Limitations include small sample sizes in several chapters, broad and heterogeneous ICD-10 groupings, possible mismatches between documentation and information available to treating clinicians, uncertainty in reference diagnoses and panel consensus, and multiple comparisons.

5 Conclusion

These exploratory findings support moving beyond aggregate accuracy summaries before clinical deployment. Where diagnostic codes are already available, ICD-10 chapter stratification offers a practical, low-cost way to examine variation across broad diagnostic subgroups. The divergence between diagnostic and safety-related performance patterns further suggests that these subgroup analyses should extend beyond diagnostic accuracy alone.

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Disclosure of Interests. The authors have no competing interests to disclose.

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